ISNAD at KnowledgeGraphEval 2026: Arabic Relation Extraction via Entity-Aware Modeling and Imbalance-Aware Resampling

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Abstract

Relation extraction (RE) is a fundamental component of knowledge graph construction, question answering, and text summarization. Although recent RE models achieve strong overall performance through the fine-tuning of pretrained language models, their aggregate performance can obscure substantial disparities across relation types, particularly under long-tailed distributions. This issue remains largely unexplored for Arabic RE. In this work, we investigate the effectiveness of class-balancing strategies for long-tail relation extraction in Arabic by fine-tuning AraBERT with entity markers on Wojood^{Relations}. Our system, submitted to the KnowledgeGraphEval 2026 shared task, achieves a Micro-F1 score of 92.73 on the official test set. Further analysis shows that errors are disproportionately concentrated among low-frequency and semantically overlapping relation types, indicating that imbalance-handling strategies developed primarily for English benchmarks do not fully address the challenges posed by the long-tailed relation distribution in Arabic.

keywords Arabic Relation Extraction, Knowledge Graphs, Wojood^{Relations}, Arabic NLP, Class Imbalance

1 Introduction

Relation Extraction (RE) is a core building block of automatic knowledge graph construction, but Arabic RE remains difficult given the language's rich morphology and the limited availability of annotated resources. Recent RE systems, built on fine-tuned pretrained language models, report strong headline yet a single aggregate metric can conceal how unevenly a model actually performs: relation types with few training examples are frequently misclassified even when overall accuracy looks high. This disparity, driven by the long-tailed distribution typical of relation extraction datasets, has received little attention in Arabic RE specifically. Existing Arabic systems, such as AREEj(Rakan Al Mraikhat et al., 2024),fine-tune pretrained models on datasets like Wojood^{Relations}(Aljabari et al., 2025b) without addressing this imbalance directly, leaving performance on minority relations largely unmeasured and unimproved.

We adopt entity markers technique for Arabic RE in our submission to the KnowledgeGraphEval 2026 shared task (Aljabari et al., 2026).

Our paper presents an empirical study of entity marking and imbalance-aware training for long-tail relation extraction in Arabic:(1) We adapt an established entity-marker representation in the sentence to represent the subject and object entities to provide the model with clearer information about the target relation. (2) We mitigate the severe class imbalance using a **WeightedRandomSampler**, applied only during training without manually modifying the original dataset. (3) We briefly summarize the system's performance across relation types and discuss the main sources of classification errors and limitations. To encourage reproducibility, full code and configuration are available on our GitHub public repository.[1]

2 Background

Arabic RE is challenging because of the linguistic variability of Arabic and the limited availability of large-scale annotated corpus. The Wojood^{Relations} corpus (Aljabari et al., 2025b) provides a large-scale resource for Arabic RE, that is built on Wojood (Jarrar et al., 2022) a nested Arabic Named Entity corpus and recognition using BERT. The Wojood^{Relations} contains 41 relation types and exhibits a highly imbalanced distribution. The task of RE is based on NER as found in *Konooz* (Hamad et al., 2025) and in *WojoodOntology* (Aljabari et al.,

[1] https://github.com/menaahmed22/ISNAD-at-KnowledgeGraphEval-2026-Arabic-RE

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2025a). Recent works in Arabic for RE tasks such as efforts in (Diaz-Garcia and Lopez, 2025) which give a comprehensive study on RE techniques. Authors in (Mashabi et al., 2026) and (Fields et al., 2024) introduce a recent survey of Large Language Models for Arabic Language and its dialects. In (Wang et al., 2023), authors conclude and discuss the research issues of domain RE.

3 Class-Balanced Relation Extraction

Our submitted system treats this task as a multi-class classification task. The system predicts one relation from a fixed set of different relation types based on a given sentence and a target pair of entities [subject, object]. The main architecture of the system consists of three main components: adding entities marks before and after subject and object in the sentence, an AraBERTV02-based classifier and imbalance-aware training using the idea of weighted sampling. The system pipeline can be summarized as Figure 1.

3.1 Data Description

We use the *Wojood*^{Relations} (Aljabari et al., 2025b) dataset, a single Arabic RE dataset constructed from *Wojood* (Jarrar et al., 2022). The dataset contains annotated semantic relations between entity pairs that cover different relation types and topics. Each sample contains a sentence, a target subject and object, and their corresponding relation label.

3.1.1 Training and validation Dataset

Thee organizers provided us with training dataset containing 17,381 instances with 41 labels. This dataset is distinguished from the original Wojood^{Relations} corpus, which has a different corpus size and split. It exhibits a high level of imbalance. The *no_relation* class contains 8,092 instances, *Location.located_in* contains 4329 instances, whereas several relation types contain fewer than 10 instances. For example, *Family.has_relative* has 2 instances, *Productivity.manufacturer_of* has 3 instances, *Organization.employs, Organization.employs, GPE.has_area and Productivity.builder_of* has 4 instances each, the complete distribution is provided in Appendix B.

In our experiments, to ensure same distribution split in the training and validation dataset for fine-tuning, We divide the provided training dataset using a stratified split with a fixed random seed and the ratio of 60 % for training (10,428 samples) and 40 % for validation (6,953 samples).

3.1.2 Testing Dataset

The test dataset provided by the shared task organization is used only for generating the submitted predictions for final evaluation without the *no_relation* label. It does not contain the target label. The model predicts the relation category from a closed set of provided relations.

3.2 Evaluation metrics

We evaluate our system by using *Micro-F1* (3) score as main evaluation metric as required in the shared task because the dataset has high imbalance among classes. Micro-F1 aggregates all the measurements' performances of all relations taking into consideration their percentage in the dataset. We also calculate accuracy, precision (1), and recall (2) for each relation to help in analyzing the behavior on the system on different relation types.

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{1}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

$$F1 - score = \frac{2PR}{P + R} \tag{3}$$

3.3 Methodology

Our submitted system is designed to treat RE as a direct multi-class classification task. Given a sentence that contains a pair of entities, the finetuned model predicts the relation between the target pair from a specified set of relation types. The architecture of the system consists of three components that are applied in sequence: (1) Adding entities marks around the subject and object in the sentence to make the model give more attention to them in the sentence during training. (2) finetuning the AraBERTv02 model with a cross-entropy loss. (3) Handling imbalance via a weighted sampler.

Our model is trained using cross-entropy loss not through binary entailment decisions in NLI-based shared task pipeline. It predicts the relation in just a single classification step.

3.4 Entity-Aware Input Representation

Entity markers are an established technique in relation extraction. We adapt this technique in our experiments, by identifying the target subject and

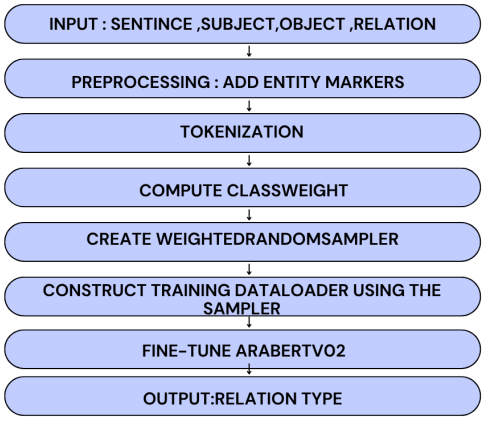


Figure 1: Overview of the submitted system

object in the sentence, providing explicit information about which entities the relation must be predicted, like below:

```
... <SUBJ> subject1 </SUBJ> ...
<OBJ> object2 </OBJ> ...
```

The marked sentence is then passed through the tokenizer of AraBERTv02. To ensure a fixed input size, sequences exceeding the maximum sequence length are truncated to a max length of 256, while shorter sequences are padded. We choose this methodology for padding and truncation as 95% of tokenized sentences after adding entity marks and applying tokenization contain below 249 tokens per sentence.

3.5 Handling Class Imbalance

One of the challenges in Wojood^{Relations} is the highly imbalanced distribution of relation classes. To solve this problem without manually modifying the dataset, we use a **WeightedRandomSampler** from `torch.utils.data` library during training. For each relation type, we calculate class weight based on the frequency of this class in the dataset using the **compute_class_weight** from `sklearn.utils.class_weight` library. The sampler then integrated into the **DataLoader** from `torch.utils.data` and is used only for constructing training batches. The evaluation is performed on the original dataset distribution. This approach increases the attendance of minority relations classes during finetuning.

3.6 AraBERTv02 Relation Classifier

We finetune different Arabic language models, but the best performance comes from the model **aubmindlab/bert-base-arabertv02** (Antoun et al., 2020) from HuggingFace library. AraBERTv02 is a BERT-based model pretrained specifically for Arabic language text and provides contextual representations for Arabic NLP tasks.

The hidden representation of the `[CLS]` tokens are used as the sentence-level representation and are passed through a linear classification layer that maps it to the 41 relation classes. The AraBERTv02 encoder and the classification layer are fine-tuned during training. The model assigns a probability score to each relation class, and the relation with the highest probability score is chosen as the prediction.

Parameter	Value
Optimizer	AdamW
Learning rate	2×10^{-5}
Train & validation batch size	32
Maximum epochs	30
Weight decay	0.01
Maximum sequence length	256
Random seed	42
Early stopping patience	5
Early stopping threshold	0.001
Model selection	Best validation F1
Sampling	WeightedRandomSampler
Sampling replacement	True

Table 1: Training configuration of the submitted system.

4 Experimental Setup

4.1 Training Configuration

Table 1 introduces the hyperparameters used for the configuration of the training process. To prevent overfitting, We evaluate the model performance at the end of each epoch using **F1_score** rather than accuracy because of the imbalance in the dataset and use early stopping technique. We save the checkpoint values after each epoch. The checkpoint with the best validation F1-score is automatically restored at the end of training.

4.2 Implementation Details

All the experiments were run on Google Colab environment (`https://colab.research.google.com/`) using **T4 GPU**. The system is implemented using PyTorch (`https://pytorch.org/`) and the HuggingFace Transformers library (`https://huggingface.co/`).We also use pandas (`https://pandas.pydata.org/`) , NumPy (`https://numpy.org/`) ,and scikit-learn (`https://scikit-learn.org/stable/`) libraries.

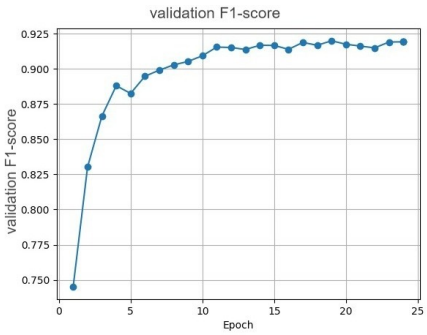


Figure 2: validation F1-score during training

System	Micro-F1 (%)
BERT baseline	88.61
LLM baseline	85.78
our system (AraBERTv02 + Markers + Sampler)	**92.73%**

Table 2: Official KnowledgeGraphEval 2026 test results.

5 Results

5.1 Official Shared Task Results

Our final submitted system achieved a Micro-F1 score of **91.92%** on validation dataset during development after 24 epochs as shown in Figure 2. The system achieved **92.73%** on the official KnowledgeGraphEval 2026 test dataset, with the predictions submitted under the name Wateen_Team. The test dataset does not contain *no_relation* class. To be sure about choosing *no_relation* label, we set probability score threshold for the output from predictions if this label is *no_relation*, the prediction is replaced with the relation label having the second-highest probability. Table 2 introduces the performance of our submitted system in comparison with the available systems reported for the shared task.

5.2 Development Experiments

To investigate the effect of the different components of our system, We performed different design choices on the training dataset during development experiments. These experiments Were conducted during system development but were not used in the official shared task submission. Based on these experiments, in our submitted system we design an architecture that combines the AraBERTv02 with entity-aware input representation and WeightedRandomSampler.

5.3 Error Analysis

Despite the strong overall performance of the submitted system 91.92% as micro f1-score , the system achieves only 74.01% macro f1-score.This gap between micro-F1 and macro-F1 shows the difficulty of predicting low-frequency relations. The imbalance in the dataset contributes in difficulty of predicting low-frequency relations. Some errors happen when the same subject-object pair is showen in different relation types across different contexts. This contextual variation makes relation prediction more challenging, particularly for semantically similar relation types. one another reason, the confusion between the semantics of the relations and the inability of the system to discover the relation and predict *no_relation*. Although weighted sampling technique increases the exposure of minority classes, the model struggle to learn reliable representation from these few examples. Figure 3 shows examples of errors from the validation dataset.

Sentence	True Relation	Predicted Relation
مقابلة مع**محمد حرب** عليان حول **قرية القالونجة المدمرة** ، محافظة عزة .	Location.lives_in	no_relation
إذ عزا **العراق الكويت** في 2 أغسطس 1990 ، ووجد الرئيس اليمني آنذاك علي عبد الله صالح نفسه في وضع لا يحسد عليه ، ,	Business.has_conflict_with	Location.has_border_with
كما ذكر بتضامن **المغربي**مع البحرين في كل القضايا التي تهم هذا البلد الشقيق ، تنفيذا للتعليمات السامية لجلالة**الملك محمد السادس** . ,	Location.lives_in	Administration.president_of

Figure 3: Representative relation classification errors.

6 Conclusion

We present our system for the Relation Extraction — KnowledgeGraphEval 2026 shared task on $Wojood^{Relations}$, presents Arabic RE as a multi-class classification task. The system combines **aubmindlab/bert-base-arabertv02** a pretrained Arabic language model with adding entity markers and a `WeightedRandomSampler` to mitigate the effects of high class imbalance. The final system achieved a Micro-F1 score of **91.92%** on validation dataset and **92.73%** on the official KnowledgeGraphEval 2026 hidden test dataset. We suggest for Future work focusing on increasing the representation of minority relation types and developing context-aware approaches to better distinguish context-dependent and relations semantically similar.

7 Acknowledgments

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Libraries' versions

-Python version: 3.12.13
-PyTorch version: 2.11.0+cu128
-Transformers version: 5.15.0
-Pandas version: 2.2.3
-Numpy version: 2.0.2
-Scikit-learn version: 1.6.1
-Matplotlib version: 3.10.0

	Relation Type	count
0	no_relation	8092
1	Location.located_in	4329
2	PartOf.geopolitical_division	1721
3	Location.lives_in	914
4	Personal.has_occupation	629
5	PartOf.subsidiary	208
6	Affiliation.member_of	173
7	Organization.has_alternate_name	169
8	Business.has_partner_with	165
9	Administration.president_of	152
10	Affiliation.employee_of	129
11	Personal.birth_place	115
12	Business.has_conflict_with	104
13	Administration.manager_of	84
14	Location.nearby	60
15	Location.headquartered_in	47
16	Administration.leader_of	36
17	Location.has_border_with	31
18	Organization.found_on	31
19	GPE.capital_of	29
20	Personal.death_date	14
21	Organization.has_propoerty	13
22	Affiliation.owner_of	13
23	GPE.has_currency	13
24	Business.has_competitor	12
25	Family.has_parent	11
26	Personal.birth_date	11
27	GPE.has_population	10
28	Organization.has_revenue	9
29	GPE.official_language	8
30	Productivity.founder_of	7
31	Affiliation.student_at	6
32	Family.has_spouse	6
33	Family.has_sibling	5
34	Productivity.builder_of	4
35	GPE.has_area	4
36	Organization.employs	4
37	Productivity.inventor_of	4
38	Organization.branch_count	4
39	Productivity.manufacturer_of	3
40	Family.has_relative	2

Table 3: The relation and instance counts

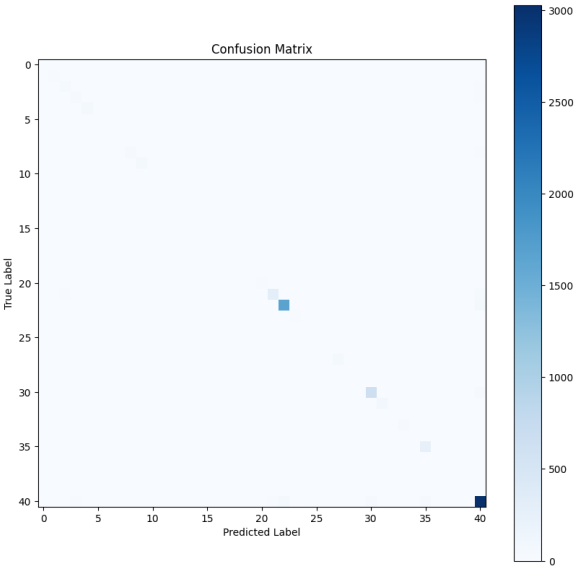


Figure 4: Confusion Matrix for validation dataset

Relation	Precision	Recall	F1	Support
Administration.leader_of	0.5882	0.7143	0.6452	14
Administration.manager_of	0.6970	0.6765	0.6866	34
Administration.president_of	0.6308	0.6721	0.6508	61
Affiliation.employee_of	0.6481	0.6731	0.6604	52
Affiliation.member_of	0.8852	0.7826	0.8308	69
Affiliation.owner_of	1.0000	0.8000	0.8889	5
Affiliation.student_at	1.0000	0.5000	0.6667	2
Business.has_competitor	0.6250	1.0000	0.7692	5
Business.has_conflict_with	0.7568	0.6667	0.7089	42
Business.has_partner_with	0.8438	0.8182	0.8308	66
Family.has_parent	0.7500	0.7500	0.7500	4
Family.has_relative	0.0000	0.0000	0.0000	1
Family.has_sibling	0.0000	0.0000	0.0000	2
Family.has_spouse	0.0000	0.0000	0.0000	2
GPE.capital_of	0.3125	0.4167	0.3571	12
GPE.has_area	1.0000	1.0000	1.0000	2
GPE.has_currency	1.0000	1.0000	1.0000	5
GPE.has_population	1.0000	1.0000	1.0000	4
GPE.official_language	1.0000	0.6667	0.8000	3
Location.has_border_with	0.8889	0.6667	0.7619	12
Location.headquartered_in	0.9375	0.7895	0.8571	19
Location.lives_in	0.8982	0.8197	0.8571	366
Location.located_in	0.9530	0.9602	0.9566	1732
Location.nearby	0.8333	0.6250	0.7143	24
Organization.branch_count	1.0000	1.0000	1.0000	1
Organization.employs	0.6667	1.0000	0.8000	2
Organization.found_on	0.9167	0.9167	0.9167	12
Organization.has_alternate_name	0.9394	0.9118	0.9254	68
Organization.has_propoerty	1.0000	1.0000	1.0000	5
Organization.has_revenue	1.0000	1.0000	1.0000	4
PartOf.geopolitical_division	0.9253	0.9360	0.9306	688
PartOf.subsidiary	0.9167	0.9277	0.9222	83
Personal.birth_date	0.6667	1.0000	0.8000	4
Personal.birth_place	0.9333	0.6087	0.7368	46
Personal.death_date	1.0000	0.6667	0.8000	6
Personal.has_occupation	0.8951	0.9484	0.9210	252
Productivity.builder_of	0.5000	1.0000	0.6667	1
Productivity.founder_of	1.0000	0.6667	0.8000	3
Productivity.inventor_of	0.6667	1.0000	0.8000	2
Productivity.manufacturer_of	0.0000	0.0000	0.0000	1
no_relation	0.9268	0.9354	0.9311	3237
Accuracy			**0.9192**	6953
Macro avg.	0.7610	0.7443	0.7401	6953
Weighted avg.	0.9194	0.9192	0.9186	6953

Table 4: Per-relation precision, recall, F1-score, and support on the validation dataset.